

Summit left front wheel speed sensor open circuit disabling ABS

Document: tsb-26-107

Area: bulletins

1 Condition

Affected Summit vehicles may illuminate the ABS warning lamp and store DTC C0035 (Left Front Wheel Speed Sensor Circuit) together with DTC U0121 (Lost Communication with ABS Control Module). With these codes active, anti-lock braking, traction control, and stability functions are disabled, though base hydraulic braking remains available. The condition is typically continuous rather than intermittent once the open circuit develops. For an overview of the sensor's role in the system, see Summit Abs Service Manual :: Wheel Speed Sensors .

2 Cause

The fault originates in the left front wheel speed sensor, part WSS-3300, whose internal winding develops an open circuit due to thermal cycling at the connector pigtail. The resulting loss of signal causes the EBCM to set C0035 and, because the module then ceases normal bus participation, the secondary code U0121. Inspect the WSS-3300 harness for chafing before condemning the sensor. Refer to Summit Abs Service Manual :: Wheel Speed Sensor Circuit Codes for detailed circuit code criteria.

3 Diagnostic Procedure

Perform diagnosis using PROC-ABS-DIAG, beginning with a scan tool capture of all stored codes and freeze-frame data. Confirm whether U0121 is a stand-alone communication fault or secondary to the sensor open circuit by checking module power, ground, and bus continuity. The communication-code logic is detailed in Summit Abs Service Manual :: System Configuration and Communication Codes , and the full module diagnostic flow is in the Summit Abs Service Manual :: ABS Module Diagnosis Procedure . Verify wheel speed sensor resistance against specification before replacement.

4 Repair Procedure

Replace the left front wheel speed sensor WSS-3300 following PROC-WSS, taking care to seat the sensor fully against the knuckle and route the harness away from the brake hose and suspension travel path. After installation, clear C0035, perform a road test to confirm wheel speed signal recovery, and verify that U0121 does not return. The complete replacement steps appear in the Summit Abs Service Manual :: Wheel Speed Sensor Replacement Procedure . General sub-system service notes are found in the Summit Abs Service Manual :: Wheel Speed Sensor Sub-System Service .